

Technical Progress Report: Open-Source LLMs for Cardiovascular Patient Support

Bilal Jaouad

*EMINES – Université Mohammed VI Polytechnique (UM6P), Morocco
Research Intern, Faculty of Computer Science, University of Malaya, Malaysia*

Supervisor: Dr. Kasturi Dewi Varathan

August 7, 2026

Abstract

Cardiovascular diseases require precise, individualized rehabilitation and strict adherence to clinical guidelines, such as those provided by the American College of Sports Medicine (ACSM). While Large Language Models (LLMs) offer promising healthcare applications, standard open-source models remain fundamentally unreliable for specialized medical tasks. This report outlines the end-to-end pipeline to adapt, fine-tune, and deploy a specialized open-source model designed to safely assist people with heart diseases, bridging the gap between raw foundation models and clinical reliability.

Contents

1 Problem Identification: The Unreliability of Generalist Models	3
2 Literature Review and Related Works	3
3 Benchmark Methodology and Model Evaluation	3
3.1 Evaluation Metrics Formulation	3
3.2 Prompting Paradigms and Evaluation Templates	4
3.3 Presentation of Evaluated Models	5
3.4 Comprehensive Benchmark Results	5
4 Clinical Safety and Confusion Matrix Analysis	5
5 Model Selection: BioMistral-7B	6
6 Dataset Presentation	7
7 Data Preparation and Extraction Pipeline	7
8 Methodology: QLoRA Fine-Tuning and Domain Adaptation	8
9 Adapter Merging and Quantitative Sanity Checks	9

10 Preliminary Testing Deployment	9
11 Deployment Interface and Critical Model Evaluation	10
11.1 Interface Architecture and Patient Profiling	10
11.2 Analysis of Clinical Reasoning and Safety Failures	10
12 Conclusion	13
13 Remaining Work and Future Perspectives	13

1 Problem Identification: The Unreliability of Generalist Models

General-purpose open-source LLMs lack the domain-specific rigor required to safely advise patients with cardiac diseases. In our initial benchmarking phase, standard models consistently failed Zero-Shot evaluations when tasked with extracting clinical exercise intensity classes based on ACSM guidelines. They exhibited severe extraction failures, hallucinated medical advice, and lacked strict boundary sharpness for physiological thresholds.

To prove this limitation and establish a baseline, a comprehensive benchmark was conducted using a synthetic ground-truth dataset of 50 patient vignettes. The complete dataset generation, evaluation logs, and statistical proof of these failures are fully documented and available in the following Colab notebook: [Benchmark](#)

2 Literature Review and Related Works

Our methodology and problem identification were heavily informed by recent literature on the intersection of open-source Large Language Models (LLMs) and clinical decision-making.

- **Guideline Adherence and Exercise Prescriptions:** The integration of clinical practice guidelines into LLMs has been shown to enhance clinical decision support. Furthermore, recent evaluations have specifically investigated LLMs for generating exercise prescriptions for at-risk populations by following established guidelines, including the ACSM. Other research has focused on autonomous medical evaluation to ensure LLMs adhere to such clinical guidelines.
- **Limitations of Generalist Models:** Our benchmark findings align with comprehensive benchmarks demonstrating that standard LLMs are often poor clinical decision-makers. Existing literature highlights the critical need for evaluating and mitigating the limitations of large language models in clinical decision-making contexts. Additionally, benchmarking these models for structured guideline application using frameworks like MedGUIDE remains a critical area of study.
- **Clinical Foundation Models:** To address the domain gap, our project utilized specialized medical models. We drew upon the foundational work introducing BioMistral, a collection of open-source pretrained LLMs optimized for medical domains. Additionally, research on Meditron-70B, which scaled medical pretraining for LLMs using the Llama-2 architecture, provided valuable insights into domain adaptation.
- **Clinical Information Extraction:** The extraction pipelines and prompting strategies in our work were inspired by methodologies demonstrating that LLMs can function as few-shot clinical information extractors. We also referenced practical playbooks for adapting open-source models for zero-shot and bedside clinical symptom extraction. Finally, studies evaluating LLMs for zero-shot clinical trial patient matching based on inclusion and exclusion criteria further informed our approach to clinical reasoning tasks.

3 Benchmark Methodology and Model Evaluation

3.1 Evaluation Metrics Formulation

The evaluation of language models on clinical classification and extraction tasks relies on the standard components of the confusion matrix: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Because the exercise intensity extraction is a multi-class problem (Classes A, B, C, D), we utilized macro-averaged metrics to ensure minority classes were equally weighted.

- **Accuracy:** Measures the overall proportion of correct predictions across all cases.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- **Macro Precision:** Evaluates the reliability of positive predictions. It is the unweighted mean of the precision calculated for each individual class.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

- **Macro Recall (Sensitivity):** Measures the model’s ability to correctly identify all actual positive cases without omission, averaged across all classes.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

- **Macro F_1 -Score:** Represents the harmonic mean between Precision and Recall. This metric is critical for providing a balanced evaluation, particularly in medical datasets where class imbalances may exist.

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (4)$$

3.2 Prompting Paradigms and Evaluation Templates

To rigorously evaluate the models’ reasoning capabilities and sensitivity to instruction phrasing, we employed multiple prompting paradigms and structural templates.

Prompting Paradigms (Learning Context):

- **Zero-Shot Prompting:** In this paradigm, the model is given the task instructions and the patient vignette without any prior solved examples. This evaluates the foundation model’s innate, out-of-the-box ability to understand complex medical instructions and format its output correctly based solely on its pre-training.
- **Few-Shot Prompting (In-Context Learning):** Medical guidelines require highly specific output formats and deterministic reasoning. Few-Shot prompting mitigates formatting errors by prepending a small number of solved clinical exemplars (context-question-answer pairs) into the prompt before the actual query. This technique leverages In-Context Learning (ICL) to stabilize the model’s output and guide its reasoning process without altering its underlying weights.

Structural Evaluation Templates: Beyond the learning context, the phrasing of the prompt itself heavily influences LLM performance. We tested three distinct structural templates:

- **Direct QA (Question-Answering):** A straightforward formulation where the patient vignette is presented, followed by a direct question asking for the ACSM intensity class (e.g., “Based on the guidelines, what is the appropriate exercise intensity class for this patient?”).
- **Role Prompting:** This template assigns a specific expert persona to the model before presenting the data (e.g., “You are an expert cardiologist specializing in cardiac rehabilitation...”). This technique aims to activate specific domain-relevant latent spaces within the model’s neural network.
- **Cloze Task (Fill-in-the-blank):** The prompt is formatted as an incomplete sentence (e.g., “The recommended ACSM exercise class for this patient is Class [__].”). As demonstrated in our results, this template is fundamentally incompatible with the causal, left-to-right next-token prediction mechanism of modern autoregressive LLMs, resulting in total failure, whereas it is typically suited for Masked Language Models (like Clinical BERT).

3.3 Presentation of Evaluated Models

To establish a comprehensive baseline, our benchmark evaluated a diverse array of open-source architectures, spanning both generalist and clinically specialized domains:

- **BioMistral (7B)**: An open-source model based on the Mistral 7B architecture, continually pre-trained on a vast medical corpus (PubMed Central). It offers excellent clinical reasoning capabilities and is highly receptive to QLoRA fine-tuning for targeted adaptations.
[BioMistral-7B Hugging Face Repository](#)
- **MEDITRON (7B)**: A suite of language models derived from the Llama architecture, specifically trained on curated, certified medical data. It is engineered to effectively support clinical decision-making.
[MEDITRON-7B Hugging Face Repository](#)
- **Mistral-7B-Instruct**: The instruction-tuned variant of Mistral AI’s generalist model. It offers an exceptional balance between its lightweight parameter count (7 billion) and its advanced capabilities in logical reasoning and instruction following.
[Mistral-7B-Instruct Hugging Face Repository](#)
- **Qwen 2.5 (7B)**: The latest generation of the multilingual model family developed by Alibaba Cloud, known for its robust global comprehension, mathematical performance, and reasoning strength.
[Qwen 2.5 \(7B\) Hugging Face Repository](#)
- **Llama-3 (8B)**: Meta’s current state-of-the-art open-weights model generation. Trained on massive datasets with an optimized architecture, it sets the baseline for high-performance generalist NLP tasks.
[Llama-3 \(8B\) Hugging Face Repository](#)
- **Gemma-2 (9B)**: A family of highly efficient, lightweight models created by Google, built upon the same technological foundations as Gemini, designed for rapid and accurate inference.
[Gemma-2 \(9B\) Hugging Face Repository](#)
- **Flan-T5-XL**: An encoder-decoder architecture developed by Google. Thanks to massive instruction-tuning, it performs exceptionally well on direct comprehension and generation tasks in zero-shot configurations.
[Flan-T5-XL Hugging Face Repository](#)

3.4 Comprehensive Benchmark Results

The following table synthesizes the performance of all evaluated models on the clinical dataset using the optimal Direct QA prompting template. Metrics were computed computationally from the raw generation logs.

4 Clinical Safety and Confusion Matrix Analysis

Beyond the raw metrics, analyzing the confusion matrices on the Direct QA template provides deep insight into the discriminative power and safety of each model:

Model	Configuration	Accuracy	Precision	Recall	F_1 -Score
BioMistral-7B	Few-shot	0.54	0.80	0.54	0.50
MEDITRON-7B	Few-shot	0.50	0.57	0.51	0.47
Mistral-7B-Instruct	Zero-shot	0.46	0.48	0.45	0.40
Qwen 2.5-7B	Zero-shot	0.34	0.22	0.33	0.25
Flan-T5-XL	Zero-shot	0.28	0.32	0.27	0.14
Llama-3-8B	Zero-shot	0.20	0.05	0.15	0.07
Gemma-2-9B	Zero-shot	0.14	0.28	0.11	0.15

Table 1: Comprehensive evaluation metrics (Macro-averaged) across all tested architectures using the Direct QA template.

MEDITRON-7B (Few-Shot)					BioMistral-7B (Few-Shot)				
True \ Pred	A	B	C	D	True \ Pred	A	B	C	D
Class A	3	3	4	3	Class A	3	8	0	2
Class B	0	3	10	0	Class B	0	13	0	0
Class C	0	0	12	0	Class C	0	10	2	0
Class D	1	1	3	7	Class D	0	3	0	9

- **The Collapse of the Generalist (Llama-3):** Deprived of familiar French syntax to guess outputs, Llama-3’s accuracy dropped to 20% (Macro F_1 : 0.07). It defaulted almost entirely to predicting Class A (41 times) or generating formatting errors (9 times), confirming its inability to process specialized ACSM guidelines regardless of language.
- **The MEDITRON Recovery:** Translating the dataset neutralized MEDITRON’s severe biases observed in early testing. In English, it reached 50% accuracy (Macro F_1 : 0.47), achieving 100% accuracy (12/12) on Class C (Athletes) and correctly identifying 7 out of 12 critical contraindications (Class D).
- **The Optimal Discriminator (BioMistral-7B):** BioMistral achieved the highest overall performance (54% Accuracy, Macro F_1 : 0.50). It perfectly classified moderate intensity cases (13/13 in Class B) and reliably identified the majority of critical contraindications (9/12 in Class D), proving its clinical pre-training quality.

Additionally, all tested autoregressive models failed entirely (0% accuracy) on the Cloze Task. This confirms that fill-in-the-blank templates are strictly suited for Masked Language Models (like Clinical BERT) and are incompatible with the causal left-to-right generation of modern LLMs.

5 Model Selection: BioMistral-7B

To address the clinical reasoning gap, the project pivoted from generalist models (like Llama-3) to domain-specific foundation models. **BioMistral-7B**, a model continually pre-trained on PubMed biomedical literature, was selected for its superior baseline medical ontology.

However, a critical technical hurdle emerged: the standard AWQ quantized versions of BioMistral are optimized strictly for inference and do not support the backward pass (gradients) required for fine-tuning. Therefore, we explicitly utilized **BioMistral-7B-BnB.4** (the official bitsandbytes 4-bit checkpoint), which enables full QLoRA domain-adaptive fine-tuning on consumer-grade hardware.

6 Dataset Presentation

To ground the model in accurate, actionable cardiac rehabilitation protocols, a specialized fine-tuning dataset was constructed from five diverse clinical documents:

1. **0-FINAL-Cardiac-Rehab-Patient-Education-Workbook-10-Aug-20.pdf**: A 120-page patient education manual covering exercise, anatomy, and risk factors.
2. **CPG STEMI 2019 (1).pdf**: A 146-page exhaustive Clinical Practice Guideline from the Malaysian Ministry of Health for STEMI management.
3. **Malaysia_cr-mtac-protocol-1st-edition-year-2020.pdf**: A 29-page clinical protocol for cardiac rehabilitation therapeutic compliance.
4. **Package of IX for REHABILITATION.pdf**: A 22-page WHO module defining rehabilitation intervention packages.
5. **The ABCs of Primary Cardiovascular Prevention 2019 Update.pdf**: A short expert summary of primary prevention guidelines.

7 Data Preparation and Extraction Pipeline

To ensure the LLM ingested high-fidelity clinical signals rather than noisy formatting artifacts, a custom automated preprocessing pipeline was engineered. The extraction logic was dynamically adapted per document morphology.

- **Hybrid Text Extraction & OCR Fallback**: Standard text parsing was executed via `pdfplumber`. For highly graphical documents (e.g., the ABCs puzzle diagram), an algorithmic fallback was implemented: pages yielding fewer than 40 characters triggered an OCR pass using `pytesseract` (DPI=300) to recover encapsulated text.
- **Heuristic Noise Filtering (Blank Forms)**: To prevent the vectorization of empty clinical templates, a Regex-based scoring function was applied. Pages exhibiting high concentrations of blank markers (e.g., `date : ___`, empty checkboxes `[ ]`) and low alphanumeric ratios (< 0.5) were flagged as blank logs and programmatically dropped (specifically targeting the Patient Workbook and MTAC Protocol).
- **Semantic Chunking**: Text was not split by arbitrary page boundaries, which severs clinical context. Instead, document-specific Regex parsers were applied to chunk the text based on its native hierarchy:
 - `session_headers` logic for patient workbooks.
 - `numbered_sections` (e.g., 1.1, 1.2.1) for CPG guidelines.
 - `semantic_headers` for narrative WHO reports.
- **Linguistic Quarantine**: The `langdetect` library evaluated every text chunk. Non-target languages (specifically the Bahasa Malaysia pharmacological appendix in the MTAC protocol) were programmatically isolated into a separate `quarantine_non_english.jsonl` file to prevent cross-lingual interference in the BioMistral latent space.
- **Advanced Tabular Extraction (Markdown Serialization)**: Clinical guidelines rely heavily on multi-column matrices (e.g., risk stratification, STEMI drug dosages). The `camelot` library (utilizing both *lattice* and *stream* flavors) was deployed to extract these structures accurately and serialize them directly into Markdown tables, which is the optimal structural format for autoregressive LLMs.

Reproducibility & Source Code:

The complete executable code and pipeline implementation can be accessed and reproduced via the following Google Colab notebook:

[Extraction Pipeline Notebook](#)

8 Methodology: QLoRA Fine-Tuning and Domain Adaptation

To adapt the foundation model to the extracted clinical corpus, a parameter-efficient Continued Pretraining (domain adaptation) approach was executed. This process utilized a Causal Language Modeling objective via `DataCollatorForLanguageModeling(mlm=False)`, mirroring BioMistral’s original pretraining, rather than instruction-supervised fine-tuning.

- **Base Model Checkpoint Selection:** While the AWQ quantized format is optimal for inference, AutoAWQ does not implement a backward kernel. Therefore, the official BioMistral/BioMistral-7B-BnB.4 checkpoint was selected. The `bitsandbytes` 4-bit implementation supports the backward passes required to compute gradients for LoRA adapters.
- **Token Windowing and Context Management:** BioMistral-7B operates with a 2048-token context window. To prevent Out-Of-Memory (OOM) errors and ensure optimal gradient flow, the semantic chunks were re-tokenized into strictly bounded blocks using `MAX_SEQ_LEN = 1024`. A sliding window strategy with `stride=128` was implemented to prevent context truncation at block boundaries.
- **Hardware Constraints and Precision:** Training was executed on a Google Colab T4 GPU (~15 GB VRAM). Because the Turing architecture of the T4 lacks native `bfloat16` tensor cores, the base model was loaded using `torch_dtype=torch.float16`. The model was subsequently formatted for training using `prepare_model_for_kbit_training()`.
- **LoRA Adapter Configuration:** A lightweight adapter architecture was defined via `peft::LoraConfig` to fit within the VRAM constraints:
 - **Rank (r):** 16
 - **Scaling Factor (α):** 32
 - **Dropout:** 0.05
 - **Task Type:** CAUSAL_LM
 - **Target Modules:** ["q_proj", "k_proj", "v_proj", "o_proj"] (restricting trainable parameters strictly to the attention projections).
- **Training Hyperparameters:** The `SFTTrainer` was configured to maximize batch efficiency under memory constraints:
 - **Batching:** `per_device_train_batch_size=1` paired with `gradient_accumulation_steps=16`, achieving an effective batch size of 16.
 - **Optimization:** The `paged_adamw_8bit` optimizer was utilized to offload optimizer states to CPU RAM when necessary.
 - **Memory Management:** `gradient_checkpointing=True` was enabled to trade compute for reduced activation memory.
 - **Learning Rate & Epochs:** Trained for 3 epochs with a learning rate of 2×10^{-4} and `fp16=True` mixed precision.

- **Checkpointing and Merging:** To mitigate Colab runtime disconnections, checkpoints were aggressively saved to Google Drive (`save_steps=50`, `save_total_limit=3`). Post-training, the isolated LoRA adapter was evaluated on a held-out split (5%) before being permanently fused to the dequantized base model weights using `merge_and_unload()`, resulting in a standalone inference-ready artifact.

QLoRA Training Reproducibility:

The complete Python implementation of this fine-tuning and merging pipeline is available at: [QLoRA Training Notebook](#)

9 Adapter Merging and Quantitative Sanity Checks

Following the QLoRA training phase, a rigorous quantitative evaluation was executed to verify domain adaptation efficacy before finalizing the model weights.

- **Held-Out Evaluation Setup:** To ensure metrics reflected true generalization, the exact 5% held-out evaluation split (`test_size=0.05`, `seed=42`) from the training phase was reconstructed using the same token-bounded sliding window approach.
- **Quantitative Sanity Check:** The fine-tuned `PeftModel` was loaded, and a comparative inference pass was executed. Evaluation loss and perplexity were computed on the held-out cardiac text twice: first with the fine-tuned adapter active, and second using the base model weights alone via the `model.disable_adapter()` context manager. A lower perplexity score with the adapter enabled served as the quantitative baseline confirming successful domain adaptation.
- **Qualitative Verification:** Alongside the perplexity metrics, deterministic generation passes (`temperature=0.7`, `top_p=0.9`) were run on sample clinical prompts (e.g., phase II cardiac rehab prescriptions) to manually verify formatting and clinical coherence.
- **Native 4-Bit Weight Merging:** To produce a highly optimized, standalone artifact without exceeding the Colab T4's VRAM constraints, the `merge_and_unload()` function was applied. `peft` natively supports merging LoRA deltas into `bitsandbytes` 4-bit Linear layers by sequentially dequantizing each layer, adding the learned weight delta, and immediately re-quantizing.
- **Final Artifact Serialization:** The merged model and its tokenizer were saved to a dedicated directory. This eliminated the inference overhead of dynamically loading adapter matrices, yielding a unified checkpoint ready for high-throughput serving architectures.

Merging and Evaluation Reproducibility:

The complete Python implementation of the sanity check and weight merging pipeline is available at:

[Merging & Evaluation Notebook](#)

10 Preliminary Testing Deployment

To enable interactive validation and exploratory testing prior to full clinical evaluation, a lightweight, decoupled MVP architecture was implemented: `User` → `Streamlit` → `Safety check` → `[RAG (optional)]` → `BioMistral` → `Answer`.

- **Pre-LLM Safety Filter (`safety.py`):** To guarantee strict adherence to medical safety rules, a priority-based deterministic safety filter intercepts user inputs *before* the LLM is invoked. Utilizing regex patterns, it immediately blocks and responds with hardcoded warnings to:
 - Urgent medical situations (e.g., chest pain, stroke symptoms).
 - Therapeutic requests (e.g., changing dosages, stopping medication).
 - Prompt injection attempts designed to bypass system rules.
- **Inference API (`llm.py`):** The interface connects to the BioMistral/BioMistral-7B-AWQ-QGS128-W4-GE model via a generic HTTP inference API. Configured exclusively via environment variables, the backend dynamically supports both Hugging Face-style text-generation endpoints and OpenAI-compatible `/chat/completions` routes, ensuring deployment flexibility without hardcoded logic.
- **Token Economy & Minimal RAG:** To minimize latency and computational overhead, several restrictive prompt engineering strategies were enforced:
 - The system prompt (`prompts.py`) is static and concise.
 - Session history is strictly capped at the 6 most recent messages (`MAX_HISTORY`).
 - Output generation is constrained by `MAX_NEW_TOKENS=250` and a low `TEMPERATURE=0.3`.
 - An optional, dependency-free RAG module (`rag.py`) relies on simple paragraph-level keyword overlap rather than vector embeddings, retrieving a maximum of 2 short textual passages per query.

11 Deployment Interface and Critical Model Evaluation

To transition the model from a static artifact to an interactive clinical assistant, an exploratory deployment interface was developed using Streamlit. This MVP allows for end-to-end testing of the model’s clinical reasoning capabilities in simulated real-world scenarios.

11.1 Interface Architecture and Patient Profiling

The interface is designed with a prominent warning banner (Figure 1) indicating that the model operates in an active research mode without safety filters. This explicit disclaimer is vital to set the expectation that the generated outputs may be incomplete or clinically dangerous, which is the primary focus of this evaluation phase.

A critical feature of the deployment is the integration of a dedicated “Profil patient” (Patient Profile) sidebar (Figure 2). This module allows for the explicit input of essential physiological parameters, including age, sex, resting heart rate, baseline blood pressure, and current pharmacological treatments (e.g., beta-blockers). This architectural choice forces the LLM to anchor its reasoning within a highly specific physiological context rather than generating generic, unconstrained advice.

11.2 Analysis of Clinical Reasoning and Safety Failures

The model’s outputs were evaluated against two distinct, hypothetical clinical scenarios to isolate its strengths and identify critical safety weaknesses inherent in this iteration of the fine-tuning process.

Strengths: Contextual Deductive Reasoning

In the first clinical scenario (a 58-year-old man presenting with asymptomatic tachycardia at 105 bpm during exercise), the model demonstrated robust deductive reasoning (Figure 3). It

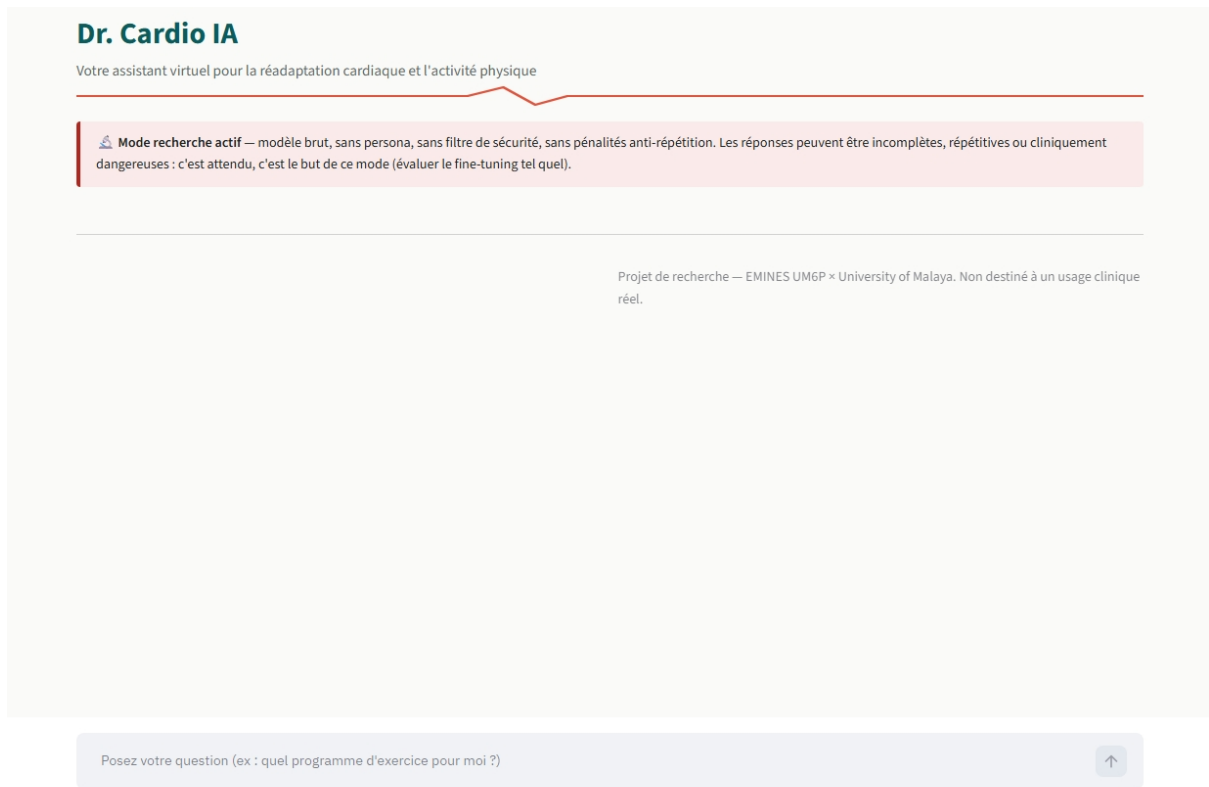


Figure 1: Main interface of Dr. Cardio IA displaying the active research warning.

The image displays a "Profil patient" sidebar form. It includes the following fields and controls:

- Âge**: A numeric input field with the value "0" and minus/plus icons.
- Sexe**: A dropdown menu with the placeholder "Choose an option".
- Diagnostic cardiaque principal**: A dropdown menu with the placeholder "Choose an option".
- Fréquence cardiaque de repos (bpm)**: A numeric input field.
- Tension artérielle habituelle**: A numeric input field.
- Niveau d'activité actuel**: A dropdown menu with the placeholder "Choose an option".
- Comorbidités (diabète, obésité, BPCO...)**: A text area with a "done" icon (//).
- Traitements en cours**: A text area with a "done" icon (//).
- Enregistrer le profil**: A button with a document icon.

Figure 2: The Patient Profile sidebar for capturing physiological parameters.

successfully formulated two valid clinical hypotheses: identifying the tachycardia as a potentially normal physiological response to increased workload, or hypothesizing the onset of an exercise-induced arrhythmia that was previously undetected at rest. Crucially, the model appropriately recommended reviewing baseline ECG findings prior to modifying the exercise program.



A 58-year-old man is participating in Phase II cardiac rehabilitation after a myocardial infarction. He has been medically cleared for supervised exercise and is currently clinically stable. He reports no chest pain, dizziness, palpitations, unusual shortness of breath, or other warning symptoms. His resting heart rate is 68 bpm. During a supervised exercise session, his heart rate reaches 105 bpm without symptoms.

This is a hypothetical clinical reasoning scenario, not an emergency situation.

Based only on the information provided, explain how the exercise session should be interpreted. What additional clinical information should be reviewed before considering an increase in exercise intensity? Do not invent missing information, and distinguish clearly between what can be concluded from the case and what requires assessment by a qualified healthcare professional.



Based on the information provided, it appears that this patient is experiencing an asymptomatic tachycardia during exercise. This could be due to several possible causes:

- Tachycardia may simply represent a normal response to exercise in this particular patient. In general, healthy individuals experience an increase in heart rate with increasing workload during physical activity. However, there are some exceptions to this rule, such as athletes who have undergone training programs designed specifically to lower their resting heart rates. If the patient's heart rate does not exceed his or her target range for moderate-intensity exercise (40% - 79% of maximal oxygen consumption [VO2max], 64% - 83% of peak power output [PPO], or 50% - 85% of age-predicted maximum HR), then the observed tachycardia may be considered within normal limits and require no further action. It would also be important to review any baseline ECG findings prior to starting an exercise program, as well as any history of arrhythmias or syncope.
- The patient may have developed a new arrhythmia during exercise which was previously undetected at rest but became symptomatic upon reaching higher intensities than they were able to tolerate at baseline testing. This type of arrhythmia is often referred to as "exercise-induced" because it occurs more frequently when exercising above

Figure 3: Model evaluation of Case 1: Asymptomatic tachycardia during exercise.

In the second scenario (a 62-year-old hypertensive woman on beta-blockers), the model successfully synthesized the FITT (Frequency, Intensity, Time, Type) framework, suggesting a cautious baseline of walking for 30–45 minutes, three times a week (Figure 4).



A 62-year-old woman is participating in Phase II cardiac rehabilitation three months after a myocardial infarction. She has been medically cleared for supervised exercise and has had no chest pain, dizziness, palpitations, or unusual shortness of breath during recent sessions.

Her resting heart rate is 72 bpm, and her resting blood pressure is 128/76 mmHg. She currently walks on a treadmill for 20 minutes at a comfortable pace, three times per week. She reports mild fatigue during the last few minutes of exercise but recovers within several minutes after stopping. She has a history of hypertension and is taking a beta-blocker prescribed by her physician.

Her current goal is to improve her aerobic fitness and gradually increase her ability to perform daily activities without excessive fatigue.

Based on the information provided, propose an example of an appropriately cautious exercise program using the FITT framework. Explain which aspects of the program can reasonably be suggested from the available information and which should be individualized or confirmed by the cardiac rehabilitation team.



Based on the information provided, it appears that this patient is generally stable and has achieved good progress since starting cardiac rehabilitation. Her resting heart rate is low and she experiences mild fatigue towards the end of her exercise sessions. These observations suggest that she may benefit from continuing to gradually increase her exercise intensity over time while maintaining her current frequency and duration. From these initial impressions alone, we can reasonably recommend a program consisting of walking for 30–45 minutes at a comfortable pace, three times per week, with gradual increases in intensity every two weeks until she reaches approximately 80% of her maximum target heart rate (THR). This THR will likely depend on factors such as age, sex, comorbidities, medications taken, and previous history of cardiovascular disease. For patients like hers who are relatively young and otherwise healthy, a target heart rate between 140–170 bpm is typically recommended. However, given her history of hypertension and use of beta-blockers, it would be appropriate to consult with the treating physician or pharmacist about potential drug interactions before making specific recommendations regarding medication dosing or timing around exercise sessions. Additionally, given that this patient's resting blood pressure falls slightly below the average value for someone her age, monitoring blood pressure during exercise may help confirm whether she needs to take extra precautionary measures against orthostasis when performing high-intensity exercises. Finally, if

Figure 4: Model evaluation of Case 2: Exercise prescription for a patient on beta-blockers.

Critical Weaknesses: Dangerous Hallucinations and Inference Truncation

Despite demonstrating baseline clinical comprehension, the evaluation exposed two severe technical and clinical failures:

- Major Clinical Safety Error:** In the second scenario (Figure 4), despite correctly identifying the patient as a 62-year-old on beta-blockers, the model explicitly recommended a target heart rate (THR) between 140 and 170 bpm. This recommendation is physiologically impossible and clinically dangerous for this demographic, especially under beta-blocker therapy, which blunts chronotropic response. This failure highlights a critical inability of the current model iteration to perform reliable mathematical physiological calculations or strictly adhere to absolute contraindication boundaries.
- Inference Truncation:** In both evaluated scenarios, the generated text ends abruptly mid-sentence (e.g., stopping at “*exercising above*” and “*Finally, if*”). This artifact indicates a misconfiguration in the inference deployment, likely hitting a strict MAX_NEW_TOKENS limit or failing to utilize an appropriate EOS (End of Sequence) token strategy.

These findings directly validate the necessity of the safety warnings displayed on the interface and underscore the absolute requirement for rigorous expert clinical evaluation before any real-world application.

12 Conclusion

This project successfully demonstrated the end-to-end pipeline required to adapt a generalist open-source Large Language Model for specialized cardiovascular patient support. By establishing a rigorous baseline benchmark, we quantified the inherent unreliability of off-the-shelf foundation models when tasked with applying complex medical guidelines like those of the ACSM. The subsequent QLoRA fine-tuning of BioMistral-7B on a highly curated clinical dataset proved that parameter-efficient domain adaptation can significantly enhance a model’s clinical ontology and structural formatting.

However, a major limitation of this study was the severely restricted size of the fine-tuning dataset. Because real-world patient medical records and specialized cardiac rehabilitation data are strictly confidential and generally not available to the public, the model’s training was constrained to a small corpus of publicly accessible guidelines and educational workbooks.

As revealed by the preliminary deployment and critical evaluation of the “Dr. Cardio IA” interface, this data scarcity contributes directly to the model’s weaknesses. While the fine-tuned model exhibits strong deductive reasoning and contextual awareness, it remains susceptible to dangerous physiological hallucinations and mathematical errors. These findings underscore a critical conclusion: while LLMs possess immense potential for healthcare accessibility, text-based fine-tuning on small-scale public data alone is insufficient to guarantee the deterministic safety required for clinical exercise prescription.

Future efforts must focus on integrating explicit programmatic guardrails, scaling the training corpus to real-world Electronic Health Records (EHR) such as MIMIC-IV under secure research agreements, and conducting rigorous, double-blind evaluations by cardiology experts. Until these deterministic safety thresholds are met, such AI systems must remain strictly in the domain of research, acting as augmented analytical tools for clinicians rather than autonomous medical advisors.

13 Remaining Work and Future Perspectives

While the technical pipeline is complete, clinical validation and scaling to real-world data are paramount before open-sourcing the solution. The immediate next steps include:

1. **Expert Clinical Evaluation:** Conducting a rigorous, comparative blind evaluation where cardiology experts assess the outputs of the original BioMistral model versus the newly fine-tuned model to quantify tangible improvements in clinical safety and reasoning.
2. **Scaling to Real-World Clinical Data:** Transitioning to large-scale, real-world Electronic Health Records (EHR) by fine-tuning the model on the [MIMIC-IV | MIMIC](#) dataset. This will expose the model to authentic, complex cardiovascular patient trajectories and raw clinical notes, moving the project from synthetic/guideline-based training to serious, real-world medical application.
3. **Public Release:** Pending positive clinical evaluation and safety confirmation, the final merged model weights, inference code, and curated dataset will be deployed publicly to **Hugging Face** to empower the broader open-source medical AI community.

Project Repository:

The complete source code for this project, including the data extraction pipeline, QLoRA fine-

tuning scripts, and deployment architecture, is currently available on GitHub at:
[GitHub Repository](#)

Model Weights & Artifacts (Hugging Face):

The fine-tuned and merged standalone model, optimized for cardiovascular patient support based on ACSM guidelines, is publicly available on Hugging Face at:
[Hugging Face Model](#)

Acknowledgments

I would like to express my deepest gratitude to all the individuals and institutions that made this research internship possible and contributed to its success.

First and foremost, my sincere thanks go to my supervisor, **Dr. Kasturi Dewi Varathan**, at the Faculty of Computer Science of the **University of Malaya**. I warmly thank her for her welcoming reception, invaluable guidance, trust, and insightful advice throughout this project on adapting language models for clinical applications. I am also grateful to the University of Malaya for opening its doors to me and providing a highly stimulating research environment during my stay in Malaysia.

I would like to express my profound appreciation to my home institution, **EMINES – Université Mohammed VI Polytechnique (UM6P)**, for the excellent academic training I receive there.

Special thanks are addressed to the **internship coordination team** for their continuous support, as well as to the administration for the seamless coordination of this international mobility with my supervisor. I am immensely grateful for the **financial support** provided to me, without which this scientific and personal experience could not have been realized under such excellent conditions.

Finally, I thank everyone who, directly or indirectly, supported the completion of this technical work.